Ofqual QAN	**600/0406/X**
Quartz ID	**1156 / 1157**
Qualification	**Level 4 Diploma in Computing / Level 4 Diploma in Computing with Business Management**

Assessment	**Databases 2.0**
Assessment Type	**Examination**
Date	**Thursday 27th February 2025**
Time	**09.00am – 12.00pm**

Examination paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

In a hospital system, data is maintained about patients, their treatments, the doctors attending to them and the wards where they are placed. A patient can undergo various treatments and each treatment might involve one or more doctors. Doctors are assigned to specific wards, with each ward hosting multiple treatments.

a) Identify **FOUR (4)** entities in the hospital database system based on the scenario provided. **4**

b) Explain what is meant by propagation constraints in a database system and identify **TWO (2)** key features of propagation constraints. **4**

c) Provide **ONE (1)** example of how propagation constraints could be applied in the hospital database. **2**

Total 10 Marks

Question 2

Revisit the hospital database.

Data integrity must be maintained when recording critical operations like admitting a patient, assigning a treatment or updating the availability of doctors.

For example, when a patient undergoes surgery, multiple updates need to happen simultaneously, such as updating patient records, logging treatment details and updating the status of the operation room.

a) Identify each property in **ACID**. Provide **ONE (1)** practical example of how each property would help ensure smooth operations for the hospital database system. **8**

b) Explain the purpose of the **ROLLBACK** command in SQL. Identify a situation where it would be used. **2**

Total 10 Marks

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Marks

Question 3

10

Scenario:

A photography studio wants to create a database to manage its clients, photography packages and bookings.

Each client can book one or more photography packages, and each photography package can be booked by multiple clients.

Draw an Entity Relationship Diagram for the photography database.

You need to identify relationships, cardinalities and label all the primary **and** foreign keys. Clearly state which entity the keys belong to.

Total 10 Marks

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Marks

Question 4

a) Flex Gym requires staff to complete the following form when members sign up to their workout plans. **8**

You are required to begin the normalisation process **and** list **EIGHT (8)** attributes that would be suitable for managing Flex Gym memberships in a database system.

Flex Gym Membership Form

Date: 15 January 2025

Member ID: M301

Member Name: Tom Johnson

Workout Plans Enrolled:

- **Strength Training** (Trainer: Luke, Duration: 6 Weeks)
- **HIIT** (Trainer: Daniel, Duration: 4 Weeks)

Personal Trainer Name: Luke

Membership Type: Gold

Membership Start Date: 01 July 2024

Payment Type: Cash/Debit or Credit Card: **CREDIT CARD**

Email Receipt: Y/N

Email Address: tom.johnson@email.com

SMS Notifications for Workout Reminders: Y/N

Additional Notes: Requires evening sessions only

b) The gym is considering switching from a spreadsheet to a database system for managing memberships and training sessions. Identify **TWO (2)** advantages of using a database system compared to the current spreadsheet. **2**

Total 10 Marks

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Marks

Question 5

a) Flex Gym wants to create a database to manage its members, workout plans, and personal trainer assignments. **6**

Members can have multiple workout plans and trainers are assigned to members for personalised coaching.

Create a CRUD matrix to show the following transactions:

Transaction 1 – Add a new workout plan to the system.

Transaction 2 – Register a new gym member.

Transaction 3 – Assign a personal trainer to an existing gym member.

Transaction 4 – Update the details of a member's workout plan, i.e., change the duration.

Transaction 5 – Remove a workout plan that is no longer offered.

Transaction 6 – Retrieve a member's personal trainer assignment history.

Transaction	Workout Plan	Member	Personal Trainer
1			
2			
3			
4			
5			
6			

b) Identify **and** explain **TWO (2)** potential issues that could arise if data integrity is not enforced when removing a workout plan that is currently in use. **4**

Total 10 Marks

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Marks

Question 6

Consider the following scenario.

A hotel management system uses a database to manage rooms, guests and reservations.

Entities:

- Room
- Guest
- Reservation

Write the appropriate SQL command to perform each of the following tasks to the room data provided below.

room

room_id	type	price_per_night (£)	is_occupied
101	Deluxe	120.00	FALSE
102	Standard	75.00	TRUE
103	Suite	137.50	FALSE

a) Create the `room` table. **5**

b) Update the `room` table to include a new column named `is_cleaned` to track whether a room is cleaned or not. **2**

c) The `price_per_night` for the room with `room_id` 102 needs to be updated to £80.00. **3**

Total 10 Marks

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Marks

Question 7

The hotel management database system contains the following tables.

room

room_id	type	price_per_night (£)	is_occupied
101	Deluxe	120.00	FALSE
102	Standard	75.00	TRUE
103	Suite	137.00	FALSE

guest

guest_id	first_name	last_name	contact_number
201	Chris	Lee	123456
202	Alice	Smith	234567
302	Aidan	Johnson	345678

reservation

reservation_id	check_in	check_out	guest_id(FK)	room_id(FK)
301	2025-12-23 09:00:00	2025-12-23 10:00:00	201	101
302	2025-04-02 12:00:00	2025-04-10 14:10:00	202	102
303	2025-10-14 08:00:00	2025-10-14 11:30:00	203	103
304	2025-09-20 19:00:00	2025-09-20 22:00:00	201	103
305	2025-09-16 11:00:00	2025-09-16 13:00:00	202	101

a) Write an SQL command to list all the reservations for rooms with the type “suite”. **3**

b) Write an SQL command to display each guest’s name, contact number and the total number of reservations for each guest who has made more than ONE (1) reservation. **4**

c) Write an SQL command to list the guest names and the room type for all guests who have reserved a room. **3**

Total 10 Marks

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Marks

Question 8

Identify **and** explain **FIVE (5)** reasons why an SME (Small or Medium Enterprise) might decide to implement a database system to replace their current manual paper-based record keeping system. **10**

Total 10 Marks

Question 9

a) A national healthcare provider relies on a central database to manage critical patient information, including medical records, treatment history and billing details. However, during a recent server failure, it was discovered that the database had no recent backups and the recovery processes were inadequate. **8**

Discuss **FOUR (4)** key areas where the lack of proper database backup and recovery processes can negatively impact the healthcare provider. For each area, include a specific example of how the failure could affect patient care or business operations.

b) A local restaurant chain is planning to expand its operations and implement a centralised database to manage inventory, orders and customer information across all its branches. The management has limited technical expertise and prefers a solution that minimises the need for managing database infrastructure. **2**

Identify a suitable cloud database model for them. Provide **ONE (1)** reason to support your answer.

Total 10 Marks

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Marks

Question 10

A global online multiplayer gaming platform is building a backend system to manage its diverse data requirements.

The platform needs to store player profiles, in-game leader boards, real-time matchmaking data and social connections between players.

To meet these needs, the platform is considering adopting **NoSQL** databases for efficient scalability and performance.

a) Define what NoSQL stands for. **1**

b) Identify **THREE (3)** NoSQL data models that could be used to store different types of data for the platform. Provide **ONE (1)** example of data that each model would handle best. **6**

c) Explain how the 3V model (Volume, Variety and Velocity) applies to the data management challenges the online multiplayer gaming platform might face when handling large-scale data. You must explain each V in this context. **3**

Total 10 Marks

End of Paper

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